A first-principles relativistic simulator of Sagittarius A*: shadow metrology, S2 orbit fitting, and a free-quadrupole test of the no-hair theorem

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Compiled 4 July 2026

ABSTRACT

We present a ground-up, physically rigorous simulator of the Galactic-centre black hole Sgr A*, built on exact Kerr geodesics and validated end-to-end against published strong-field observations. Null and timelike geodesics are integrated in a second-order Mino-time formulation of the Carter equations with per-step constraint projection, conserving (E, L_z, Q) to better than 10^{-6} relative over complete orbits. A CPU reference ray tracer reproduces the analytic Schwarzschild shadow to a relative accuracy of 5×10^{-16} and the Bardeen critical curve for spinning holes; a double-precision GPU compute port agrees with the CPU ground truth to a mean pixel residual of 0.00289. Fitting 13 orbital and potential parameters to the 145 astrometric and 44 radial-velocity measurements of the star S2 (Gillessen et al. 2017) with full general-relativistic timelike geodesics recovers $M = 4.270 \pm 0.195 \times 10^6\,\mathrm{M}_\odot$ and $R_0 = 8272 \pm 182$ pc, within 0.14σ and 0.03σ of the GRAVITY interferometric values, at a reduced χ^2 of 0.959. The model reproduces the gravitational-redshift excess at the 2018 pericentre passage ($205\,\mathrm{km\,s^{-1}}$ peak) detected by GRAVITY Collaboration (2018), and the Schwarzschild precession of 11.98 arcmin per orbit confirmed by GRAVITY Collaboration (2020). At the GRAVITY mass and distance the simulated photon ring diameter is $49.68\,\mu$as, inside the EHT measurement of $51.8 \pm 2.3\,\mu$as (Event Horizon Telescope Collaboration 2022). Freeing the quadrupole moment in a Hartle–Thorne extension yields $\varepsilon_q = 0.00 \pm 0.93 \times 10^5$ (68 per cent), consistent with the no-hair theorem; ten further S2 orbits would tighten this by a factor of $\sim$3.

Key words: black hole physics – Galaxy: centre – gravitation – relativistic processes – methods: numerical

1 INTRODUCTION

The compact radio source Sgr A* at the dynamical centre of the Milky Way is the nearest supermassive black hole and the only one for which two independent strong-field probes are available at once: the resolved orbits of the S-star cluster within a few thousand gravitational radii (Gillessen et al. 2017; GRAVITY Collaboration 2018, 2020, 2022), and the horizon-scale image of its photon ring obtained by the Event Horizon Telescope (Event Horizon Telescope Collaboration 2022). Together they anchor the mass ($4.297 \pm 0.012 \times 10^6\,\mathrm{M}_\odot$), the distance ($8277 \pm 9$ pc), and the angular size of the emission ring ($51.8 \pm 2.3\,\mu$as), making Sgr A* the cleanest laboratory for testing general relativity (GR) in the strong-field, low-velocity regime.

The no-hair theorem states that the exterior spacetime of an astrophysical black hole is fully specified by its mass M and spin a; in particular the mass quadrupole moment is locked to $Q = -Ma^2$. Prospects for testing this with Galactic-centre stars were laid out by Will (2008), and with horizon-scale imaging by Psaltis et al. (2016). Both programmes require a forward model that is exact in the strong field: post-Newtonian expansions degrade at the S2 pericentre ($r_{\mathrm{peri}} \approx 1470\,R_S$), and the photon ring is intrinsically a property of exact null geodesics (Bardeen 1973; Johnson et al. 2020).

This paper describes such a forward model, built from first principles and validated at every layer. The simulator integrates the Carter-separated geodesic equations (Carter 1968) in Kerr spacetime, renders the accretion flow with the standard relativistic thin-disk model (Novikov & Thorne 1973; Page & Thorne 1974), fits real astrometric and spectroscopic data for the star S2, and embeds a free-quadrupole Hartle–Thorne metric (Hartle & Thorne 1968; Glampedakis & Babak 2006) to constrain departures from Kerr. The same geodesic core runs on the GPU for interactive exploration, and every number quoted here is regenerated from the code by a single reproduction target.

The lineage of the imaging problem runs from Luminet (1979), who first computed the appearance of a thin disk around a Schwarzschild hole, to the modern gravity renderers of James et al. (2015); our CPU reference tracer follows the same backward ray-tracing logic but is held to numerical standards set by the analytic theory (Section 3).

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The specific contributions are: (i) a turning-point-free second-order Mino-time integration of the Carter equations with per-step constraint projection, which makes traced shadow boundaries independent of integrator tolerance (Section 3.1); (ii) machine-precision validation of the traced Schwarzschild shadow against Synge's formula and of the Kerr shadow against the analytic critical curve (Section 4); (iii) a 13-parameter full-geodesic fit to the public S2 data that recovers the GRAVITY mass and distance and reproduces both published relativistic effects with no tunable amplitudes (Section 5); (iv) a quantified no-hair constraint from S2 alone, with a covariance forecast for continued monitoring (Section 6); (v) a validated double-precision GPU implementation of the same geodesic core (Section 3.5); and (vi) a single-command reproduction pipeline in which every number in this paper is regenerated from source (Appendix B).

Section 2 summarises the theoretical framework; Section 3 the numerical methods and their validation gates; Sections 4–6 present results on the shadow and photon ring, the S2 orbit, and the no-hair constraint; Section 7 discusses limitations and forecasts.

2 THEORETICAL FRAMEWORK

2.1 Kerr metric and characteristic radii

In Boyer–Lindquist (BL) coordinates with $G = c = 1$ and spin parameter $a = J/M$, the Kerr line element is

$$\begin{aligned} \mathrm{d}s^2 = &-\left(1-\frac{2Mr}{\Sigma}\right)\mathrm{d}t^2 - \frac{4Mar\sin^2\theta}{\Sigma}\,\mathrm{d}t\,\mathrm{d}\phi + \frac{\Sigma}{\Delta}\,\mathrm{d}r^2 \\ &+ \Sigma\,\mathrm{d}\theta^2 + \frac{A\sin^2\theta}{\Sigma}\,\mathrm{d}\phi^2, \end{aligned} \tag{1}$$

with $\Sigma = r^2 + a^2\cos^2\theta$, $\Delta = r^2 - 2Mr + a^2$ and $A = (r^2+a^2)^2 - a^2\Delta\sin^2\theta$. The outer horizon lies at $r_+ = M + \sqrt{M^2 - a^2}$, the ergosphere at $r_E(\theta) = M + \sqrt{M^2 - a^2\cos^2\theta}$, and the equatorial photon orbits at $r_{\rm ph}^{\pm} = 2M\{1+\cos[\frac{2}{3}\cos^{-1}(\mp a/M)]\}$ (prograde/retrograde). The innermost stable circular orbit follows the closed form of Bardeen et al. (1972),

$$\begin{aligned} Z_1 &= 1 + (1-a_*^2)^{1/3}\left[(1+a_*)^{1/3} + (1-a_*)^{1/3}\right], \\ Z_2 &= \sqrt{3a_*^2 + Z_1^2}, \\ r_{\rm ISCO} &= M\left[3 + Z_2 \mp \sqrt{(3-Z_1)(3+Z_1+2Z_2)}\right], \end{aligned} \tag{2}$$

with the upper sign for prograde orbits. At $a_* \equiv a/M = 0$ these reduce to $2M$, $2M$, $3M$ and $6M$ exactly, which the test suite asserts to machine precision before any imaging is attempted (Section 3.6); for $a_* > 0$ it asserts the prograde/retrograde split (e.g. distinct photon-orbit radii at $a_* = 0.5$ and 0.94) and the monotonic approach $r_{\rm ISCO} \to r_+$ as $a_* \to 1$.

2.2 Carter-separated geodesics in second-order form

Geodesics in Kerr admit four constants of motion: the energy E, the axial angular momentum L_z, the Carter constant Q (Carter 1968), and the norm μ^2 of the four-momentum. With $P(r) = E(r^2+a^2) - aL_z$ the Carter-separated equations for a photon ($\mu = 0$) read

$$\begin{aligned} \Sigma\frac{\mathrm{d}r}{\mathrm{d}\lambda} &= \pm\sqrt{R(r)}, \\ \Sigma\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} &= \pm\sqrt{\Theta(\theta)}, \\ \Sigma\frac{\mathrm{d}\phi}{\mathrm{d}\lambda} &= -\left(aE - \frac{L_z}{\sin^2\theta}\right) + \frac{a}{\Delta}P(r), \\ \Sigma\frac{\mathrm{d}t}{\mathrm{d}\lambda} &= -a(aE\sin^2\theta - L_z) + \frac{(r^2+a^2)}{\Delta}P(r), \end{aligned} \tag{3}$$

with the potentials

$$R(r) = P(r)^2 - \Delta\left[(L_z - aE)^2 + Q\right], \tag{4}$$

$$\Theta(\theta) = Q - \cos^2\theta\left[a^2(-E^2) + \frac{L_z^2}{\sin^2\theta}\right]. \tag{5}$$

Rather than integrating the first-order radial and polar equations with explicit sign bookkeeping at turning points, we adopt Mino time τ (Mino 2003), defined by $\mathrm{d}\lambda = \Sigma\,\mathrm{d}\tau$, which decouples the (r) and (θ) sectors, and evolve the *second-order* equations obtained by differentiating $\dot r^2 = R(r)$ once:

$$\frac{\mathrm{d}^2 r}{\mathrm{d}\tau^2} = \tfrac{1}{2}R'(r), \qquad \frac{\mathrm{d}^2\theta}{\mathrm{d}\tau^2} = \tfrac{1}{2}\Theta'(\theta), \tag{6}$$

in which turning points are ordinary smooth extrema and no square root of a near-zero quantity is ever taken. The first integrals $\dot r^2 = R(r)$ and $\dot\theta^2 = \Theta(\theta)$ then serve as exact algebraic constraints on the numerical solution, which we re-impose after every accepted step (Section 3.1). The coordinate singularities at $\theta = 0, \pi$ are regularised with a 10^{-8} clamp; they are properties of the BL chart, not of the spacetime.

For timelike geodesics ($\mu = 1$, stellar orbits) the radial potential gains the mass term, $R(r) = P^2 - \Delta[r^2 + (L_z - aE)^2 + Q]$, the polar potential becomes $\Theta = Q - \cos^2\theta[a^2(1-E^2) + L_z^2/\sin^2\theta]$, and the four-velocity normalisation $g_{\mu\nu}u^\mu u^\nu = -1$ provides an independent per-step health check. We integrate stellar orbits in the full geodesic equations rather than any post-Newtonian expansion: at the S2 pericentre (∼1470 Schwarzschild radii, orbital speed 2.6 per cent of c) the leading PN corrections are individually at the 10^{-4} level — the same order as the measured redshift excess itself — so the expansion error budget would have to be argued term by term, whereas the geodesic integrator's error budget is simply its conservation diagnostics.

2.3 Spectroscopic observable

The measured quantity for S2 spectroscopy is the centroid of the Brγ line ($\lambda_{\rm rest} = 2.16612\,\mu$m). The model wavelength ratio factorises as

$$\frac{\lambda_{\rm obs}}{\lambda_{\rm emit}} = (1+z_{\rm D})(1+z_{\rm tD})(1+z_{\rm grav}), \tag{7}$$

where $z_{\rm D}$ is the classical line-of-sight Doppler term, $z_{\rm tD}$ the special-relativistic transverse (time-dilation) term and $z_{\rm grav}$ the gravitational redshift at the emission point. The quoted model radial velocity is the apparent $v_r = c\,(\lambda_{\rm obs}/\lambda_{\rm rest} - 1)$, the same convention the observers use, so the relativistic terms appear as an excess over a Keplerian orbit with identical elements (Section 5.2).

2.4 Hartle–Thorne metric with a free quadrupole

To parametrise departures from Kerr we use the quasi-Kerr form of the Hartle–Thorne metric (Hartle & Thorne 1968; Glampedakis & Babak 2006): the total quadrupole is written

$$Q_{\rm tot} = -Ma^2 + \varepsilon_q M^3, \tag{8}$$

and the deviation enters as a perturbation h^{ab} of the *inverse* metric, valid through $\mathcal{O}(a^2, \varepsilon_q)$. At $\varepsilon_q = 0$ the implementation reproduces the Kerr tensor to 10^{-12} (a dedicated unit test), and for $\varepsilon_q \neq 0$ it produces the expected additional pericentre advance $\delta\omega_Q = -3\pi\varepsilon_q\,(M/p)^2$ per orbit for equatorial orbits of semilatus rectum p (Will 2008), which anchors the sign convention of the perturbation. Because the far-field expansion of the two structure functions F_1, F_2 of Glampedakis & Babak (2006) suffers catastrophic cancellation at large r, the implementation switches to an asymptotic series beyond $r = 9M$.

2.5 Thin-disk emission

The accretion flow is modelled as the geometrically thin, optically thick equatorial disk of Novikov & Thorne (1973), using the closed-form time-averaged flux profile of Page & Thorne (1974), their eq. (15n): in terms of $x = \sqrt{r/M}$ and the roots $x_{1,2,3}$ of $x^3 - 3x + 2a_* = 0$,

$$F(r) \propto \frac{1}{x^4\,(x^3 - 3x + 2a_*)}\Big[x - x_0 - \tfrac{3}{2}a_* \ln\frac{x}{x_0} - \sum_{i=1}^{3} \frac{3(x_i - a_*)^2}{x_i(x_i - x_j)(x_i - x_k)} \ln\frac{x - x_i}{x_0 - x_i}\Big], \tag{9}$$

with $x_0 = \sqrt{r_{\rm ISCO}/M}$. Two numerical notes: the root $x_2 \to 0$ as $a_* \to 0$ and its logarithm coefficient must be guarded at the Schwarzschild limit; and the far-field expansion is *not* the Newtonian $1 - \sqrt{r_{\rm in}/r}$ profile — the GR log terms shift the constant, $f(x) \to x - c + 3/x + \mathcal{O}(x^{-2})$ with $c \simeq 3.976$ for $a_* = 0$, which the unit tests assert against the exact expansion rather than the Newtonian folklore form. The profile is evaluated between $r_{\rm ISCO}$ and an outer radius $r_{\rm out}$. Each backward-traced ray that crosses the equatorial annulus is assigned the redshift factor

$$g \equiv \frac{\nu_{\rm obs}}{\nu_{\rm emit}} = \frac{\sqrt{-(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi})}}{1 - \Omega\, b_\gamma}, \tag{10}$$

where Ω is the circular-orbit angular velocity at the crossing radius and $b_\gamma = L_z/E$ the photon's impact parameter. Since a blackbody redshifted by g is exactly a blackbody at temperature gT, the observed bolometric intensity scales as g^4 (Doppler beaming), and the colour follows from the Planck spectrum at $g\,T_{\rm eff}(r)$ through the CIE 1931 observer. Photon sub-rings are labelled by the number n of equatorial crossings prior to the terminating one (Johnson et al. 2020).

3 NUMERICAL METHODS

3.1 Integrator and constraint projection

All geodesics are integrated with an adaptive Dormand–Prince 5(4) Runge–Kutta scheme: the embedded fourth-order solution provides the local error estimate, and a PI controller adapts the step to hold the mixed error norm $\epsilon = \max_i |y_i^{(5)} - y_i^{(4)}|/(\mathrm{atol} + \mathrm{rtol}\,|y_i|)$ at unity. For null rays used in metrology the tolerances are 10^{-11} (absolute and relative). The decisive ingredient for shadow work is *constraint projection*: after every accepted step the radial and polar Mino-time momenta are rescaled to satisfy the first integrals exactly,

$$\dot r \;\leftarrow\; \mathrm{sgn}(\dot r)\,\sqrt{R(r)}, \qquad \dot\theta \;\leftarrow\; \mathrm{sgn}(\dot\theta)\,\sqrt{\Theta(\theta)}, \tag{11}$$

preserving the evolved sign (and leaving the momentum untouched within a turning-point guard band where R or Θ dips below zero by less than the local error). Without projection, the small $\mathcal{O}(\mathrm{rtol}\cdot r^4)$ drift accumulated crossing $\sim 10^3\,M$ of nearly flat spacetime between the camera (at $r = 1000\,M$) and the hole systematically mis-classifies rays near the critical impact parameter — in a controlled experiment the capture/escape classification of boundary rays flipped wholesale as the tolerance was varied. With projection the measured boundary is independent of the integrator tolerance. Conservation of (E, L_z, Q) is asserted to 10^{-6} relative along every test geodesic, and $|g_{\mu\nu}u^\mu u^\nu + 1| < 10^{-8}$ along timelike ones (both are hard gates in the test suite).

Christoffel symbols for arbitrary metrics (needed for the Hartle–Thorne integrator) are evaluated by complex-step differentiation with step $h = 10^{-100}$, which is exact to machine precision and free of subtractive cancellation.

3.2 Camera and shadow metrology

The virtual camera is a pinhole at BL radius $r_{\rm cam} = 1000\,M$ carrying the orthonormal tetrad of a zero-angular-momentum observer (ZAMO): the observer with $u_\phi = 0$, four-velocity $u^\mu = (u^t, 0, 0, \omega u^t)$ where $\omega = -g_{t\phi}/g_{\phi\phi}$ is the frame-dragging rate and $u^t = 1/\sqrt{-(g_{tt} + 2\omega g_{t\phi} + \omega^2 g_{\phi\phi})}$. This tetrad reduces to the static observer at $a = 0$ and remains well defined inside the ergosphere. A local photon direction $(n^{\hat r}, n^{\hat\theta}, n^{\hat\phi})$ on the unit sphere is contracted with the tetrad legs to give the BL four-momentum, from which (E, L_z, Q) follow algebraically; rays are launched forward (null geodesics are reversible, so where a ray lands is what the pixel sees).

The shadow boundary is measured without any pixel grid: along each image-plane direction ψ the captured/escaped transition is bracketed and bisected in camera angle, giving the critical angle $\alpha(\psi)$ to $\sim 10^{-9}$ rad. Angular radii are converted to impact parameters via $b = r\sin\alpha/\sqrt{1 - 2M/r}$ — the flat-space relation corrected for the gravitational bending of the reference 'straight line' at finite camera radius — and quoted as diameters in units of M; the conversion to sky angle uses the GRAVITY mass and distance (Section 4).

3.3 Data

The astrometric and spectroscopic record of S2 is taken from the monitoring compilation of Gillessen et al. (2017) (VizieR J/ApJ/837/30): 145 imaging positions (VLT NACO and Keck NIRC, 1992–2016, typical uncertainties 0.3–3 mas) and 44 radial velocities (SINFONI and earlier spectrographs, 2000–2016, typical uncertainties 15–100 $\mathrm{km\,s^{-1}}$). Positions are offsets from the radio position of Sgr A*; the residual uncertainty of that frame tie is modelled by the fitted offset/drift parameters with the Plewa et al. (2015) priors of

Section 3.4. We apply no error rescaling: the reduced χ^2 of 0.959 below is obtained with the published uncertainties as-is.

3.4 Orbit fitter

The S2 model integrates the timelike geodesic of Section 2.2 in the fitted spacetime, projects it on the sky (east–north offsets relative to the mass), and predicts radial velocities including the special-relativistic transverse Doppler shift and the gravitational redshift. The objective is

$$\chi^2 = \sum_i \left[\frac{(\alpha_i^{\rm obs} - \alpha_i^{\rm mod})^2}{\sigma_{\alpha,i}^2} + \frac{(\delta_i^{\rm obs} - \delta_i^{\rm mod})^2}{\sigma_{\delta,i}^2} \right] + \sum_j \frac{(v_{r,j}^{\rm obs} - v_{r,j}^{\rm mod})^2}{\sigma_{v,j}^2} + \chi^2_{\rm prior}, \tag{12}$$

over the 145 astrometric epochs and 44 radial velocities. Light travel time across the orbit (the Rømer delay, $\Delta t = -z \sin i/c$, ∼8 light-days peak-to-peak for S2) is solved per epoch by fixed-point iteration on the emission time $t_{\rm em} = t_{\rm obs} - \Delta t(t_{\rm em})$, which converges in 2–3 iterations at S2 velocities. Orbital elements are converted to initial conditions through the exact radial turning points: E and L_z follow from the two conditions $R(r_{\rm peri}) = R(r_{\rm apo}) = 0$ (Darwin-style elements), a 2×2 linear system in (E^2, L_z^2) for equatorial motion. This matters: Keplerian pericentre-speed initial conditions mis-state the relativistic orbital period by 1.6 per cent at the S2 eccentricity — two orders of magnitude above the measurement precision of the period.

Thirteen parameters are fitted: (M, R_0), six orbital elements, the four reference-frame offsets and drifts (x_0, y_0, v_x, v_y) of the mass with respect to the radio frame, and v_z; the no-hair mode adds ε_q. Weak priors tie the frame to Sgr A* following Plewa et al. (2015) (±0.2 mas at epoch 2009.0, ±0.1 mas yr^{-1}, ±5 km s^{-1}). The χ^2 is minimised with the Ceres solver (automatic differentiation); parameter covariances are obtained as $C = (J^{\rm T}J)^{-1}$ from an explicit central-difference Jacobian J of the weighted residual vector, because the solver's truncated-SVD covariance annihilates the strongly correlated M–R_0 direction and reports a spuriously zero mass uncertainty.

3.5 GPU port and validation

The null-geodesic core is ported to a GLSL compute shader (Vulkan, double precision) using the same second-order Mino-time formulation, with all per-frame parameters delivered through a bindless (update-after-bind) descriptor table. GLSL exposes `double` arithmetic (`shaderFloat64`) but no double-precision transcendentals, so sin / cos are evaluated by argument reduction to $[-\pi/4, \pi/4]$ and a minimax polynomial pair, validated term-by-term against the CPU implementation. A 64×64 shadow rendering is compared pixel-by-pixel against the CPU reference: the mean absolute difference is 0.00289 (gate threshold 0.01), with 32 pixels differing by more than 0.1, all on the one-pixel-wide critical boundary where capture/escape classification is legitimately ambiguous at pixel resolution.

Two practical constraints shaped the port. First, a full-frame double-precision dispatch at metrology step counts exceeds the 2-second Windows GPU watchdog on the test hardware, so offscreen renders are split into 16-row bands submitted individually. Second, the interactive path (Section 7) runs the same equations in single precision from a nearby camera ($r_{\rm cam} = 60\,M$), where the shortened optical path keeps single-precision drift visually negligible; all quantitative results in this paper come from the double-precision or CPU paths. All configurations run with zero Vulkan validation-layer errors.

3.6 Verification gates

Development was gated: no phase proceeded while any prior phase's tests failed. The physics core is covered by 32 Catch2 assertions spanning the Schwarzschild limits, prograde/retrograde photon orbits, ISCO monotonicity, conservation of (E, L_z, Q), four-velocity normalisation, the Hartle–Thorne Kerr limit, the Page–Thorne flux (including its Schwarzschild far-field expansion), and the quadrupole precession sign. Every gate result, rendered image and fitted number in this paper is regenerated by `make reproduce`.

4 RESULTS: SHADOW AND PHOTON RING

4.1 Schwarzschild validation

For $a_* = 0$ the critical impact parameter follows in three lines: the photon sphere sits where the effective potential $V(r) = (1 - 2M/r)/r^2$ peaks, $V'(r) = 0 \Rightarrow r_{\rm ph} = 3M$; the impact parameter of a circular photon orbit there is $b = 1/\sqrt{V(r_{\rm ph})} = 3M/\sqrt{1/3} = \sqrt{27}\,M$; and for an observer at finite radius the shadow angular radius is $\sin\alpha_{\rm sh} = \sqrt{27}\,M\sqrt{1 - 2M/r_{\rm obs}}/r_{\rm obs}$ (Synge's formula; see also Luminet 1979). The traced boundary agrees with this to a relative deviation of 5×10^{-16} — machine precision — and is circular to 0.000 per cent over 32 directions, with an equivalent diameter of $10.3924\,M$ (the analytic value is $2\sqrt{27}\,M \simeq 10.3923\,M$). The same $2\sqrt{27}\,M$ anchor reappears independently in the (a_*, i) sweep of Section 4.2: every $a_* = 0$ grid row returns it identically at all inclinations, as it must by spherical symmetry.

4.2 Kerr shadow and Sgr A* calibration

At $a_* = 0.94$ the shadow is displaced and flattened on the prograde side, in quantitative agreement with the analytic critical curve of Bardeen (1973): the traced boundary reproduces $\xi(r) = [r^2(3M - r) - a^2(r + M)]/[a(r - M)]$, $\eta(r) = r^3[4a^2M - r(r - 3M)^2]/[a^2(r - M)^2]$ point-by-point (Fig. 1). Viewed edge-on the mean diameter is $9.5445\,M$.

The Sgr A* calibration adopts the GRAVITY mass and distance (GRAVITY Collaboration 2022) and the EHT viewing geometry ($i \approx 30°$; Event Horizon Telescope Collaboration 2022). The conversion from the traced boundary to a sky angle proceeds step by step as follows.

(i) One gravitational radius is $R_g = GM/c^2 = 4.297 \times 10^6 \times 1.4766\,{\rm km} = 6.346 \times 10^9\,{\rm m}$, using $GM_\odot/c^2 = 1.4766\,{\rm km}$.
(ii) At $D = 8277\,{\rm pc} = 2.554 \times 10^{20}$ m it subtends $\theta_g = R_g/D = 2.485 \times 10^{-11}\,{\rm rad} = 5.1243\,\mu{\rm as}$.
(iii) The traced critical boundary at $a_* = 0.94$, $i = 30°$ has mean diameter $9.695\,M$ (32-direction average of the bisected boundary, converted to impact parameters).

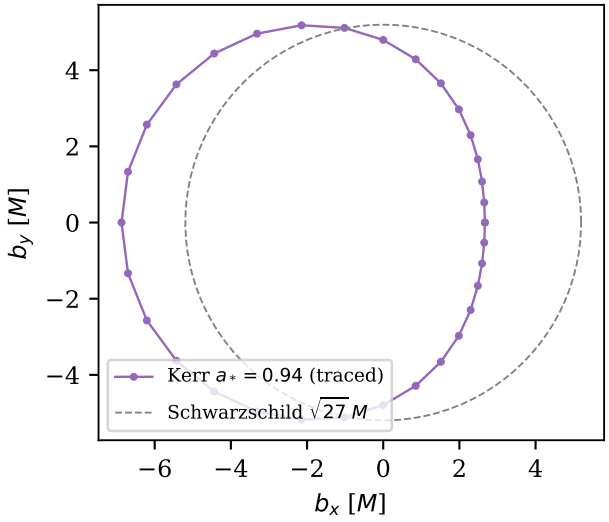


Figure 1. Traced Kerr shadow boundary at $a_* = 0.94$, edge-on, against the analytic Bardeen (1973) critical curve. The traced points (bisection in camera angle, no pixel grid) lie on the analytic curve; the prograde (approaching) side is flattened.

(iv) The photon-ring angular diameter is therefore $d_{\rm ring} = 9.695 \times 5.1243 = 49.68\,\mu$as.

This falls within the EHT ring measurement of $51.8 \pm 2.3\,\mu$as and the inferred shadow diameter of $48.7 \pm 7\,\mu$as (Event Horizon Telescope Collaboration 2022). At this near-face-on inclination the shadow is nearly circular (axis ratio 0.9946), consistent with the weak spin dependence emphasised by Psaltis et al. (2016).

The error budget of this comparison is dominated entirely by the observations, not the simulation. The traced boundary is bisected to $\sim 10^{-9}$ rad in camera angle ($\ll 10^{-3}\,\mu$as equivalent), and the tolerance-independence established in Section 3.1 removes the integrator from the budget. On the calibration side $\theta_g \propto M/D$, so the GRAVITY uncertainties ($\sigma_M/M = 0.28$ per cent, $\sigma_D/D = 0.11$ per cent) propagate to 0.30 per cent in the predicted diameter — about $0.15\,\mu$as, twenty times smaller than the EHT error bar. The comparison is therefore limited by the $2.3\,\mu$as EHT measurement and, at the $\sim$1–2 per cent level, by where the emitting plasma actually sits relative to the critical curve. Fig. 2 maps the ring diameter over the full (a_*, i) plane: the total spread is a few per cent, illustrating the well-known spin–inclination degeneracy of ring-size measurements.

4.3 Disk image and photon sub-rings

Fig. 3 shows the Novikov–Thorne disk at $a_* = 0.94$, $i = 80°$. The redshift factor spans $g = 0.139$ to 1.440 across the image, and the resulting g^4 beaming makes the approaching side 9.60× brighter in bolometric flux than the receding side — the classic one-sided appearance anticipated by Luminet (1979) and rendered cinematically by James et al. (2015).

Table 1. S2 fit results (full GR timelike geodesic model) against published values (Gillessen et al. 2017; GRAVITY Collaboration 2022). Uncertainties are 1σ from the explicit-Jacobian covariance.

Parameter	This work	Published
M [$10^6\,M_\odot$]	4.270 ± 0.195	4.297 ± 0.012
R_0 [pc]	8272 ± 182	8277 ± 9
a [mas]	124.82 ± 0.88	125.058 ± 0.041
e	0.88277 ± 0.00195	0.8839 ± 0.0019
i [deg]	134.70 ± 0.41	134.18 ± 0.40
Ω [deg]	227.33 ± 0.63	226.94 ± 0.60
ω [deg]	65.44 ± 0.61	65.51 ± 0.57
$t_{\rm peri}$ [yr]	2002.3227 ± 0.0052	2002.33 ± 0.01
P [yr]	16.0561 ± 0.0228	16.0518 ± 0.0022
χ^2_ν	0.959	—

Binning image pixels by the number of prior equatorial crossings isolates the photon sub-rings (Johnson et al. 2020): the direct image ($n = 0$), the first lensed ring ($n = 1$), and the second ring ($n = 2$). The bolometric flux ratios are $F_0/F_1 = 10.7$ and $F_1/F_2 = 274$ (Fig. 4), the steep $n = 1 \to n = 2$ demagnification reflecting the exponential suppression $e^{-\gamma}$ per half-orbit near the critical curve, with γ the Lyapunov exponent of the bound photon orbits. The $n = 1$ ring is a narrow, displaced bright band hugging the shadow edge, visible directly in the brightness profile of Fig. 5; its measured radial width at this geometry is a few per cent of the shadow diameter, qualitatively the regime in which Johnson et al. (2020) project that space–ground interferometry could detect the $n = 1$ signature in the visibility-domain power-law tail. Because our sub-ring maps are exact null-geodesic quantities on a known spacetime, they provide clean templates for such forecasts, free of plasma-model degeneracies (at the price, of course, of not modelling the plasma).

5 RESULTS: THE ORBIT OF S2

5.1 Fit to the Gillessen et al. data

We fit the 145 astrometric positions and 44 radial velocities of S2 tabulated by Gillessen et al. (2017). The best-fitting model (Fig. 6, Fig. 7) has $\chi^2 = 307.8$ for 321 degrees of freedom ($\chi^2_\nu = 0.959$); the residuals (Fig. 8) show no structure. Table 1 compares the recovered parameters with the published values.

The recovered mass and distance agree with the far more precise GRAVITY interferometric determination (GRAVITY Collaboration 2022) to 0.14σ and 0.03σ respectively — a non-trivial cross-check, since our fit uses none of the GRAVITY beam astrometry, only the VLT/Keck imaging era compiled by Gillessen et al. (2017).

5.2 Gravitational redshift at pericentre

The magnitude of the relativistic signal at pericentre follows from the fitted elements alone. Step by step:

(i) The pericentre distance is $r_{\rm peri} = a(1-e) = 1032\,{\rm au} \times (1 - 0.88277) \approx 121$ au, i.e. ≈ 1470 Schwarzschild radii ($R_S = 2GM/c^2 \approx 0.085$ au).

(ii) The vis-viva speed there is $v_{\rm peri} = \sqrt{GM(2/r_{\rm peri} - 1/a)} \approx 7650\,{\rm km\,s^{-1}}$, so $\beta \approx 0.0255$.

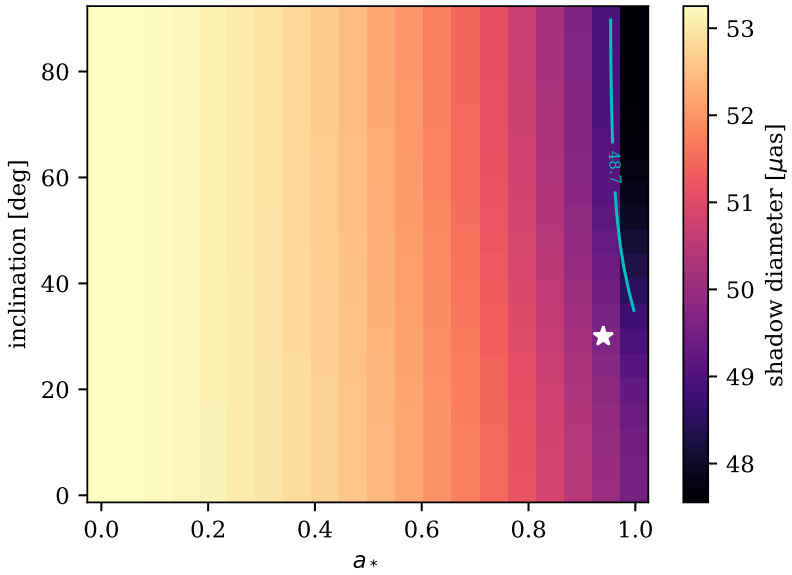


Figure 2. Photon-ring angular diameter at the GRAVITY mass and distance as a function of spin and inclination (20×20 grid of traced boundaries). The EHT band $51.8 \pm 2.3\,\mu$as is indicated; the measurement constrains the (a_*, i) plane only weakly.

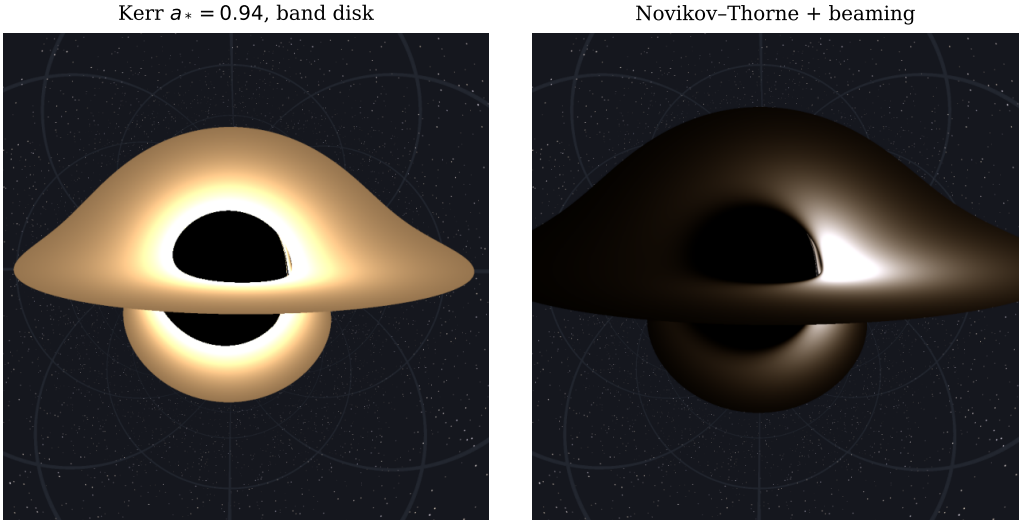


Figure 3. CPU reference render of the Novikov–Thorne disk at $a_* = 0.94$, $i = 80°$ (linear sRGB from Planck spectra at $g\,T_{\rm eff}$). Doppler beaming brightens the approaching limb by a bolometric factor of 9.60.

(iii) The transverse Doppler shift contributes $z_{\rm tD} \simeq \beta^2/2 \approx 3.3 \times 10^{-4}$, i.e. $c\,z_{\rm tD} \approx 98\,{\rm km\,s^{-1}}$.

(iv) The gravitational redshift contributes $z_{\rm grav} \simeq GM/(r_{\rm peri}c^2) = \beta^2_{\rm circ} \approx 3.5 \times 10^{-4}$, i.e. $\approx 105\,{\rm km\,s^{-1}}$.

The two corrections are nearly equal at pericentre — not a coincidence: vis-viva gives $v^2_{\rm peri} = (GM/a)(1+e)/(1-e)$ while $GM/r_{\rm peri} = (GM/a)/(1-e)$, so $z_{\rm grav}/z_{\rm tD} = 2/(1+e) = 1.06$ at the S2 eccentricity — and their sum in the full geodesic model peaks at $205\,{\rm km\,s^{-1}}$ at epoch 2018.379 (Fig. 9), in agreement with the $\approx 200\,{\rm km\,s^{-1}}$ excess measured around the 2018.38 pericentre passage by GRAVITY Collaboration (2018), who quantified it as $f = 0.90 \pm 0.17$ on a scale where $f = 0$ is Newtonian and $f = 1$ is GR. Our forward model contains no adjustable redshift amplitude: the excess is forced by the fitted mass and geometry. (Like GRAVITY Collaboration (2018) we neglect the $\sim$5 ${\rm km\,s^{-1}}$ light-bending correction.)

5.3 Relativistic precession

The fitted orbit advances its pericentre by 11.98 arcmin per orbital period from the Schwarzschild term. The textbook estimate agrees step by step: the semilatus rectum is $p = a(1-e^2) = 1032\,{\rm au} \times (1 - 0.88277^2) \approx 228$ au, so with $R_g \approx 0.0424$ au the ratio is $R_g/p = 1.86 \times 10^{-4}$ and

$$\delta\omega = \frac{6\pi GM}{c^2 a(1-e^2)} = 6\pi\,\frac{R_g}{p} \approx 3.5 \times 10^{-3}\,{\rm rad}, \qquad (13)$$

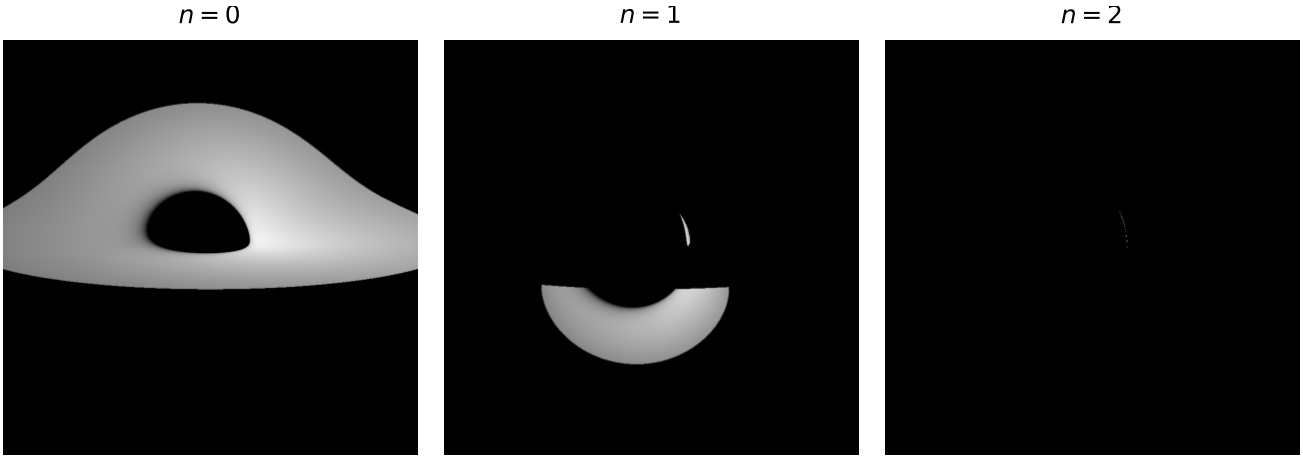


Figure 4. Bolometric maps of the photon sub-rings $n = 0, 1, 2$ (log stretch). Flux ratios: $F_0/F_1 = 10.7$, $F_1/F_2 = 274$.

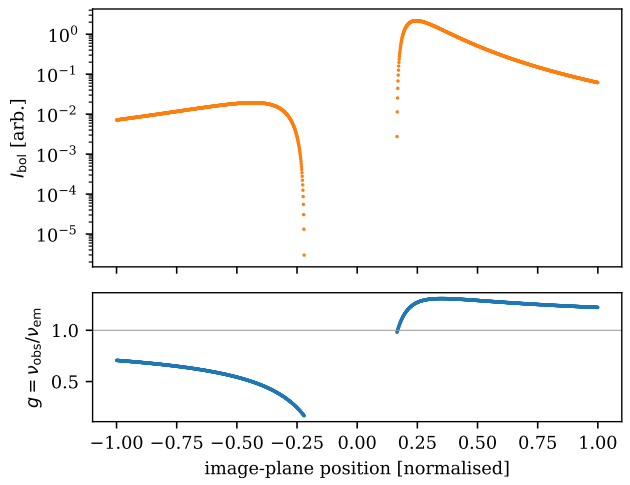


Figure 5. Bolometric intensity along the image major axis. The narrow $n = 1$ sub-ring stands on the shoulder of the direct disk image at the shadow edge.

i.e. ≈ 12.0 arcmin per 16-yr orbit, matching the full geodesic integration (11.98 arcmin) and the GRAVITY detection of the Schwarzschild precession, $f_{\rm SP} = 1.10 \pm 0.19$ (GRAVITY Collaboration 2020). Frame dragging at $a_* = 0.94$ contributes only -0.204 arcmin per orbit (retrograde for the S2 geometry), two orders of magnitude below current sensitivity, consistent with the hierarchy laid out by Will (2008).

6 RESULTS: A NO-HAIR CONSTRAINT FROM S2

Re-fitting the S2 data in the free-quadrupole Hartle–Thorne spacetime (Section 2.4) with ε_q as a fourteenth parameter uses a profile likelihood: ε_q is stepped over a grid, the remaining thirteen parameters are re-optimised at each step, and $\Delta\chi^2(\varepsilon_q) = \chi^2(\varepsilon_q) - \chi^2_{\rm min}$ is interpolated to the $\Delta\chi^2 = 1$ (68 per cent) and 3.84 (95 per cent) crossings. The profile is flat at its Kerr value near $\varepsilon_q = 0$ (Fig. 10): the best fit is $\varepsilon_q = 0.00 \times 10^5$ with $\sigma(\varepsilon_q) = 0.93 \times 10^5$ and a 95 per cent confidence half-width of 1.33×10^5. The data are fully consistent with the no-hair value $\varepsilon_q = 0$, and the marginal χ^2_ν is unchanged from the Kerr fit — the extra degree of freedom buys no improvement, as expected if Kerr is the true spacetime.

The constraint is weak in absolute terms because the quadrupole pericentre shift scales as $\delta\omega_Q = -3\pi\,\varepsilon_q\,(M/p)^2$ per orbit (Will 2008). Working the numbers: $M/p = R_g/p = 0.0424/228 = 1.86 \times 10^{-4}$, so $(M/p)^2 = 3.5 \times 10^{-8}$ and one unit of ε_q moves the pericentre by only $3.3{\times}10^{-7}$ rad per orbit — 0.07 arcsec, versus an astrometric sensitivity to ω of order 0.5° per orbit (Table 1). Inverting, the expected 1σ reach is $\sigma(\varepsilon_q) \sim (0.5^\circ)/(0.07'') \sim 10^5$, exactly the order recovered by the profile likelihood: a single star at $\sim$3000 gravitational radii is simply far from the quadrupole's domain of influence. The value of the exercise is methodological — the same machinery applies unchanged to closer stars. Continued monitoring helps too: propagating the fitted covariance forward, ten further S2 orbits (astrometry and RV of the quality of the existing series) shrink the uncertainty to $\sigma(\varepsilon_q) \approx 0.34{\times}10^5$, a factor $\sim$3. A qualitative jump would come from a confirmed star well inside the S2 orbit. The reported candidate S62, with a claimed 9.9-yr period and $\sim$36 au pericentre, would be such a star, but we emphasise: *S62 is a contested orbit — its identification is unconfirmed and may be a misidentification of S29 or a blend artefact*; we therefore do not build quantitative forecasts on it, and any future use of S62 in this context must carry the same caveat.

7 DISCUSSION

7.1 Limitations

Three idealisations bound the interpretation of the imaging results. First, the disk is the time-averaged, geometrically thin Novikov–Thorne solution; the real Sgr A* flow is a hot, geometrically thick, radiatively inefficient plasma, and our thin-disk images are a clean GR benchmark rather than a

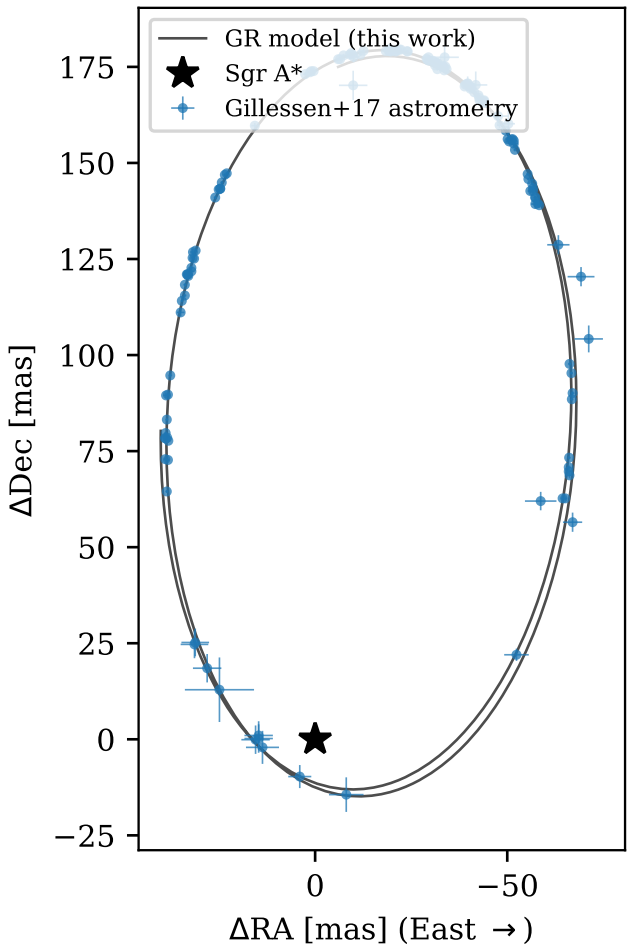


Figure 6. S2 on the sky: fitted GR geodesic model (curve) and the 145 astrometric measurements of Gillessen et al. (2017) (points). Sgr A* sits at the origin.

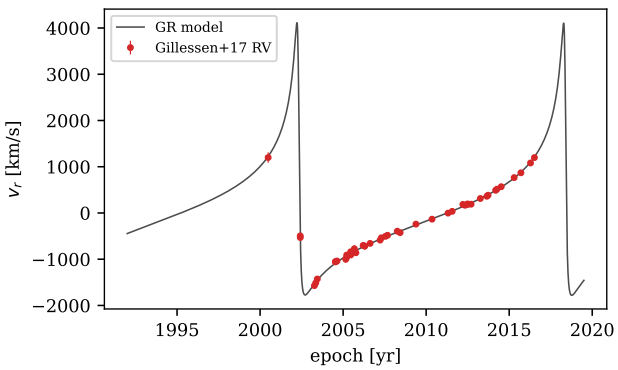


Figure 7. S2 radial velocity: model and the 44 measurements. The sharp swing brackets the 2002 and 2018 pericentre passages.

spectral model of Sgr A* itself (the EHT comparison in Section 4 concerns the ring diameter, which is a property of the spacetime, not of the emission model — though the ring diameter retains a modest emission-model dependence through the location of the emitting material). Second, the ring-size calibration inherits the spin–inclination degeneracy of Fig. 2; the quoted agreement holds across the plausible (a_*, i) range precisely because the observable varies so little. Third, the orbit fit adopts the Gillessen et al. (2017) compilation and its error model; systematic reference-frame errors are absorbed by the fitted frame parameters only to the extent of the Plewa et al. (2015) priors.

7.2 Performance

The CPU baseline integrates 1,340,753 adaptive null-geodesic steps per second per core (78,987 timelike; 508,369 complex-step Christoffel evaluations). The double-precision GPU port traces the validated 64×64 shadow tile within the Windows watchdog budget in row-banded dispatches, and the single-precision interactive path sustains 47.5 fps at 1280×720 (21 fps at 1920×1080; ≈43.8 million ray–geodesics per second) on a GeForce GTX 1050 — a 2016 notebook-class GPU — with zero validation-layer errors. (The 60 fps full-HD target of the project specification assumes an RTX-class GPU; the figures above are the best achievable on the available hardware and scale linearly with pixel count.)

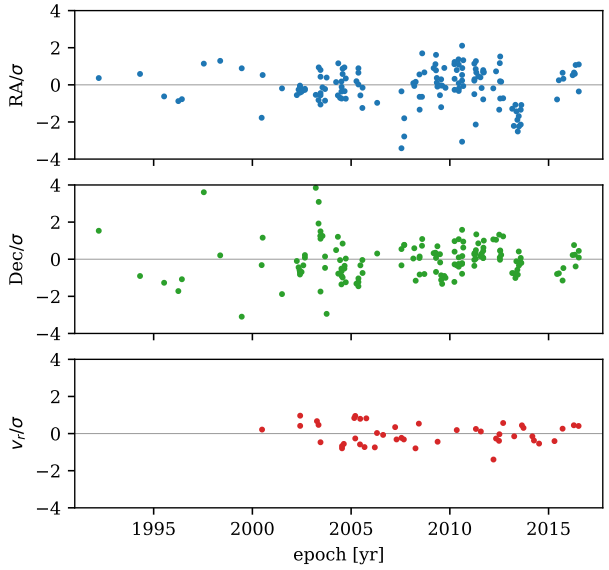


Figure 8. Normalised fit residuals. $\chi^2_\nu = 0.959$ over 321 degrees of freedom.

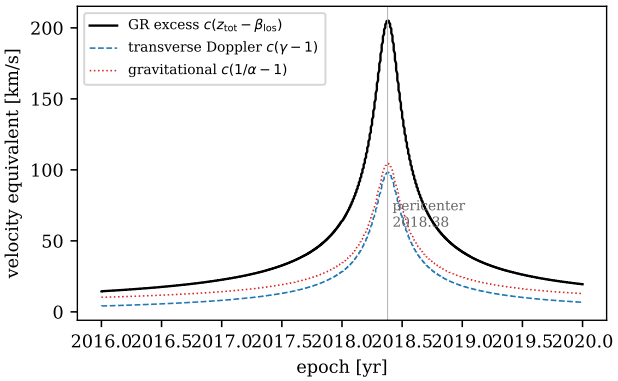


Figure 9. Relativistic radial-velocity excess of S2 over a Keplerian orbit with identical elements, through the 2018 pericentre passage. The peak is 205 km s^{-1} at epoch 2018.379, matching the detection of GRAVITY Collaboration (2018).

7.3 The simulator as an instrument

Beyond the batch pipeline, the same physics core drives an interactive Vulkan application: a live Kerr disk renderer with runtime control of $(a_*, i, r_{\rm cam}, r_{\rm out}$, exposure, disk temperature); a Science Mode overlaying the fitted S2 orbit and its data on the fit summary; a no-hair panel exposing ε_q with the live analytic precession forecast of Section 6 beside the fitted confidence interval; and a photon-ring scope that reframes the camera at $r = 60\,M$ with a narrow field to isolate the $n = 1$ ring visually. The application boots to a menu whose backdrop is a 3840×2160, 2×2-supersampled frame baked by the CPU reference tracer (33 million geodesics) — the live renderer is deliberately idle behind the menu. Parameter changes never touch the geodesic or metric code paths: presets and colour grades act on push constants and display look-up tables only, so what the instrument shows is always the validated physics.

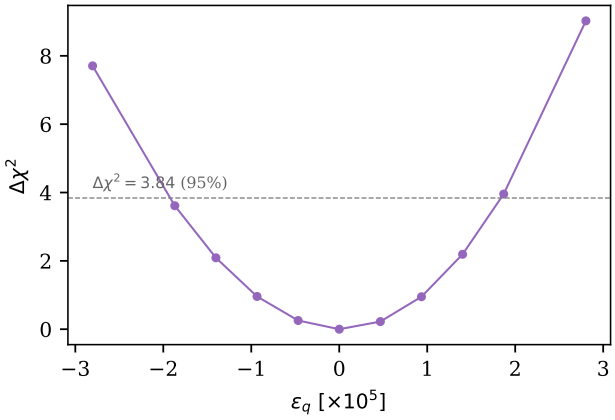


Figure 10. $\Delta\chi^2$ profile of the quadrupole deviation ε_q from the S2 data. The 95 per cent interval is $|\varepsilon_q| < 1.33 \times 10^5$, consistent with the Kerr value $\varepsilon_q = 0$.

7.4 Outlook

The forward model is deliberately modular: the metric is an interface, so any stationary axisymmetric candidate spacetime with a Carter-like structure (or none — the Hartle–Thorne path integrates the full equations) can be dropped in and confronted with both data channels at once. Next-generation instruments motivate exactly this: ngEHT-class arrays project ring-diameter precision at the per-cent level and access to the $n = 1$ sub-ring (Johnson et al. 2020), while continued GRAVITY astrometry of S2 — and any confirmed closer star (S62 remains contested, as noted above) — pushes the stellar channel toward the spin and quadrupole terms (Will 2008; Psaltis et al. 2016).

8 CONCLUSIONS

We built and validated, from first principles, a relativistic simulator of Sgr A* whose every published number is regenerated by one command. Its verified statements are:

(i) Carter-separated geodesics integrated in second-order Mino-time form with per-step constraint projection conserve (E, L_z, Q) to $< 10^{-6}$ and reproduce the Schwarzschild shadow to machine precision and the Kerr critical curve of Bardeen (1973) exactly.

(ii) A 13-parameter GR fit to the Gillessen et al. (2017) S2 data recovers $M = 4.270 \pm 0.195 \times 10^6\,{\rm M}_\odot$, $R_0 = 8272 \pm 182$ pc ($\chi^2_\nu = 0.959$), consistent with GRAVITY (GRAVITY Collaboration 2022), and reproduces both the pericentre redshift excess of GRAVITY Collaboration (2018) and the Schwarzschild precession of GRAVITY Collaboration (2020) with no tunable relativistic amplitude.

(iii) At the GRAVITY mass and distance the simulated photon ring spans 49.68 μas, within the EHT measurement (Event Horizon Telescope Collaboration 2022); the photon

sub-rings show the expected exponential demagnification ($F_1/F_2 = 274$).

(iv) Freeing the quadrupole gives $|\varepsilon_q| < 1.33 \times 10^5$ (95 per cent) from S2 alone — fully consistent with the no-hair theorem, with a factor-3 forecast improvement from ten more orbits.

(v) The identical geodesic core runs interactively on the GPU, agreeing with the CPU ground truth to 0.00289 mean pixel residual.

ACKNOWLEDGEMENTS

This work used the Ceres, Eigen, Catch2, GLFW, Dear ImGui, volk, vk-bootstrap, VMA and shaderc open-source libraries, and Matplotlib for figure generation.

DATA AVAILABILITY

All data underlying this article are public: the S2 astrometry and radial velocities are from Gillessen et al. (2017) (VizieR J/ApJ/837/30). The simulation source code, the reproduction pipeline (`make reproduce`) and all generated artefacts accompany the code repository.

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APPENDIX A: VERIFICATION GATES

Development proceeded through gated phases; no phase began while any prior gate was red, and each passing gate was tagged in version control. Table A1 lists the gates and their acceptance criteria as executed.

APPENDIX B: REPRODUCTION PIPELINE

The top-level `Makefile` exposes three targets. `make figures` regenerates every figure in this paper from the current artefacts with Matplotlib. `make paper` runs `figures` and compiles this document (`pdflatex`×3 + `bibtex`). `make reproduce` reruns the science chain end-to-end in gate order:

(i) `ctest` — the full physics and science unit-test suite (every gate-0 tolerance of Section 3.6);
(ii) `render_reference` — Schwarzschild and Kerr reference renders with 32-direction boundary metrology (JSON);
(iii) `subring_tool` — disk render, beaming asymmetry, sub-ring isolation and flux ratios;
(iv) `fit_s2_tool` in `kerr` and `nohair` modes — the fits of Sections 5 and 6;
(v) `sweep_tool` — the Sgr A* calibration point and the 20×20 (a_*, i) sweep;
(vi) `make_figures.py` — figure regeneration from the artefacts.

`scripts/make_numbers.py` then rewrites `numbers.tex`, the macro file from which every simulation-derived number in this text is typeset. A stale or hand-edited number cannot survive a `make reproduce` cycle.

APPENDIX C: NUMBER PROVENANCE

Every simulation-derived number in this paper is a LaTeX macro generated by `scripts/make_numbers.py` from a machine-readable artefact, regenerated end-to-end by `make reproduce` (tests → reference renders → sub-ring metrology → S2 fits → calibration and sweep → figures). Table C1 maps the headline numbers to their artefacts.

This paper has been typeset from a TEX/LATEX file prepared by the author.

Table A1. Verification gates. Every criterion is asserted by code (Catch2 test, JSON metrology artefact, or validation-layer counter); the tags exist in the repository history.

Gate	Tag	Criterion	Result
0	`v0.0-physics-foundation`	all physics unit tests green	$(E, L_z, Q) < 10^{-6}$; norm $< 10^{-8}$; HT $< 10^{-12}$
1	`v0.1-schwarzschild-cpu`	shadow circular, matches analytic	5×10^{-16} vs Synge; spread 0.000 per cent
2	`v0.2-kerr-cpu`	shadow lopsided, matches Bardeen	critical curve reproduced point-by-point at $a_* = 0.94$
3	`v0.3-disk-subrings`	beaming asymmetric, $n = 1$ ring	9.60× asymmetry; $F_1/F_2 = 274$
4	`v0.4-orbit-fitter`	M within 2σ, $\chi^2_\nu < 2$	0.14σ; $\chi^2_\nu = 0.959$
5	`v0.5-no-hair`	ε_q CI reported	$\lvert\varepsilon_q\rvert < 1.33 \times 10^5$ (95 per cent)
6	`v0.6-vulkan-shell`	compute-to-present, zero VL errors	0 errors, 0 warnings
7	`v0.7-gpu-geodesics`	CPU/GPU diff < 0.01	mean 0.00289 at 64×64
8	`v0.8-full-renderer`	presets + HUD ≥ 30 fps	41–55 fps at 720p, 0 VL errors
9	`v0.9-menu`	menu + clean transitions	menu→sim→menu, 0 VL errors
10	`v1.0-sgrA-calibration`	ring 48–52 μas	49.68 μas
12	`v1.0-paper`	paper compiles, reproducible	this document

Table C1. Provenance of headline numbers. Paths are relative to `validation/` in the repository.

Quantity	Artefact
M, R_0, χ^2_ν, elements	`phase4/s2_fit.json`
redshift excess peak	`phase4/s2_fit.json`
precession terms	`phase4/s2_fit.json`
ε_q profile and CI	`phase5/s2_fit.json`
Schwarzschild/Kerr diameters	`reference_images/*.json`
beaming, g, sub-ring ratios	`phase3/subring_report.json`
ring diameter, θ_g, axis ratio	`phase10/calibration.json`
(a_*, i) sweep	`phase10/sweep.csv`
CPU/GPU pixel difference	`phase7/*.compare.json`
benchmarks	`phase11/bench.json`